Indian Institute of Information Technology, Surat
Curriculum: B.Tech. (ECE) Electronics & Communication Engineering

Regular Entry to the Course

- Candidates who have cleared JEE (Main) are eligible
- Admissions managed by CSAB and JoSAA

Semester-1

Sr. No	Course	Code	Credit	Teaching Scheme			Examination Scheme			Total
				L	T	P	L	T	P	
1	Fundamentals of Computers & Programming	CS 101	4	3	0	2	100	0	50	150
2	Digital Design	EC 102	4	3	0	2	100	0	50	150
3	Engineering Physics	AS 103	4	3	0	2	100	0	50	150
4	Engineering Mathematics	AS 104	3	3	0	0	100	0	0	100
5	ICT Workshop - I	EC 105	2	0	0	4	0	0	100	100
6	Engineering Circuit Analysis	EC 106	2	2	0	0	50	0	0	50
7	Indian Constitution	HM 107	2	2	0	0	50	0	0	50
	Total		21	16	0	10	500	0	250	750
	Total Contact Hours per Week			26						

Semester-2

Sr. No	Course	Code	Credit	Teaching Scheme			Examination Scheme			Total
				L	T	P	L	T	P	
1	Data Structure and Algorithm	CS 201	4	3	0	2	100	0	50	150
2	Electronic Devices and Circuits	EC 202	4	3	0	2	100	0	50	150
3	Computer Architecture & Organisation	CS 203	3	3	0	0	100	0	0	100
4	Discrete Mathematics	AS 204	3	3	0	0	100	0	0	100
5	ICT Workshop - II	CS 205	2	0	0	4	0	0	100	100
6	Yoga Practice	HM 206	2	2	0	0	50	0	0	50
	Total		18	14	0	8	450	50	200	700
	Total Contact Hours per Week			22						

Lateral Entry – 1

- Candidates who have cleared JEE (Main) with a rank preferably within 50000 (appropriate concession in the rank will be applicable to SC and ST candidates), and earned a minimum 39 credits in first year B.Tech offered by any institute, with ECE curriculum equivalent to that of IIIT-Surat.

OR

- Candidates having Diploma in appropriate branch after 12th and have cleared JEE (Main) with a rank preferably within 50000 (appropriate concession in the rank will be applicable to SC and ST candidates).
- Admission applications submitted along with NoC from the previous institute will be scrutinized for eligibility.
- Candidates will be admitted only if the current strength does not exceed the total sanctioned strength in each course.

Semester-3

Sr. No	Course	Code	Credit	Teaching Scheme			Examination Scheme			Total
				L	T	P	L	T	P	
1	Operating System	CS 301	4	3	0	2	100	0	50	150
2	Electrical Networks	EC 302	3	3	0	0	100	0	0	100
3	Electronic Circuits	EC 303	4	3	0	2	100	0	50	150
4	Signals & Systems	EC 304	4	3	0	2	100	0	50	150
5	Probability & Statistical Analysis	AS 305	3	3	0	0	100	0	0	100
6	Economics & Business Management	HM 306	2	2	0	0	50	0	0	50
	Total		20	17	0	6	550	0	150	700
	Total Contact Hours per Week			23						

Semester-4

Sr. No	Course	Code	Credit	Teaching Scheme			Examination Scheme			Total
				L	T	P	L	T	P	
1	Control Systems	EC 401	4	3	0	2	100	0	50	150
2	Computer Networks	CS 402	4	3	0	2	100	0	50	150
3	Analog Circuits	EC 403	4	3	0	2	100	0	50	150
4	Communication Systems	EC 404	4	3	0	2	100	0	50	150
5	Embedded Systems	EC 405	4	3	0	2	100	0	50	150
6	Electromagnetics	EC 406	3	3	0	0	100	0	0	100
	Total		23	18	0	10	600	0	250	850
	Total Contact Hours per Week			28						

Semester-5

Sr. No	Course	Code	Credit	Teaching Scheme			Examination Scheme			Total
				L	T	P	L	T	P	
1	Wireless Communication	EC 501	4	3	0	2	100	0	50	150
2	Nanoscale Device Engineering	EC 502	3	3	0	0	100	0	0	100
3	Image Processing & Computer Vision	EC 503	4	3	0	2	100	0	50	150
4	Cloud Computing & Big Data Infrastructure	CS 504	4	3	0	2	100	0	50	150
5	Innovation & Entrepreneurship	HM 505	2	2	0	0	50	0	0	50
6	Elective - 1*	-	3	-	0	-	-	0	-	100
	Total		20	-	0	-	-	0	-	700
	Total Contact Hours per Week			-						
	* Electives will be offered either as 3 – 0 – 0 or 2 – 0 – 2 scheme.									

Semester-6

Sr. No	Course	Code	Credit	Teaching Scheme			Examination Scheme			Total
				L	T	P	L	T	P	
1	Machine Learning	CS 601	4	3	0	2	100	0	50	150
2	Digital VLSI Design	EC 602	4	3	0	2	100	0	50	150
3	Web Engineering	CS 603	4	3	0	2	100	0	50	150
4	Mini Project	EC 604	3	0	0	6	0	0	100	100
5	Elective - 2*	-	3	-	0	-	-	0	-	100
6	Elective - 3*	-	3	-	0	-	-	0	-	100
	Total		21	-	0	-	-	0	-	750
	Total Contact Hours per Week			-						
	* Electives will be offered either as 3 – 0 – 0 or 2 – 0 – 2 scheme.									

Electives				
5thSemester			6thSemester	
E-1	• Mathematical Optimization (EC511) • Hardware Description Languages (EC513) • Analog VLSI Design (EC514) • Probabilistic Graphical Model (CS512) • Fuzzy & Neural Networks (CS514) • Object Oriented Programming (CS516)		E-2	• Sensors & Instrumentation (EC621) • Microwave communication systems (EC623) • MEMS (EC624) • Introduction to Game Design (CS622) • Human Computer Interface (CS623) • Data Mining and Data Warehousing (CS624) • Wireless Sensor Networks (CS625)
			E-3	• Information Theory & Coding(EC631) • Digital Signal Processors (EC632) • Robotics and Humanoids (EC633) • Augmented & Virtual Reality (CS631) • Advanced Internet Technologies (CS634) • Bioinformatics (CS636)

<u>Exit Option – 1</u>

- A candidate willing to exit after successful completion of three years course will be given a degree as ***Bachelor of Science (BS) in Electronics and Communication Engineering***
- Candidates will be allowed to leave only if the current strength doesn't fall below 95% of the total sanctioned strength in each course.

Semester-7

Sr. No	Course	Code	Credit	Teaching Scheme			Examination Scheme			Total
				L	T	P	L	T	P	
1	Artificial Intelligence	CS 701	4	3	0	2	100	0	50	150
2	Electric Vehicle Technology	EC 702	3	3	0	0	100	0	0	100
3	Elective - 4*	-	3	-	0	-	-	0	-	100
4	Elective - 5*	-	3	-	0	-	-	0	-	100
5	Elective - 6*	-	3	-	0	-	-	0	-	100
	Total		16	-	0	-	-	0	-	550
	Total Contact Hours per Week			-						
	* Electives will be offered either as 3 – 0 – 0 or 2 – 0 – 2 scheme.									

Electives					
7th Semester					
E-4	• Satellite Communication (EC 742) • Speech Processing (EC 743) • VLSI Testing & Verification (EC 744) • Block-Chain and Ledger (CS741) • Cyber Security (CS742) • Deep Learning (CS743)	E-5	• Biomedical Electronic Systems (EC 752) • VLSI Packaging Aspects (EC 753) • Power electronics & renewable energy (EC 754) • Information Retrieval (CS751) • Computer Ethics & Public Policy (CS753) • Ethical Hacking (CS754)	E-6	• Internet of Things (EC 761) • RF Circuit Design (EC 762) • Electronic System Design (EC 763) • Distributed Computing Systems (CS761) • Mobile Computing (CS762) • Social Network Analysis (CS763)

Semester-8

Sr. No	Course	Code	Credit	Teaching Scheme			Examination Scheme			Total
				L	T	P	L	T	P	
1	B.Tech. Project	EC 801	20	0	0	40	0	0	800	800
	Total		20	0	0	40	0	0	800	800
	Total Contact Hours per Week			40						

Exit Option - 2

- A candidate leaving after successful completion of fourth year will be given ***B.Tech. degree in Electronics and Communication Engineering***

Elective Courses

- The elective subjects are offered based on current trends in the relevant industry.
- Electives can be offered either in 3-0-0 or in 2-0-2 scheme.
- The elective courses are listed below:

Information Theory & Coding	Hardware Description Languages	Fuzzy Logic & Neural Networks
Speech Processing	Sensors & Instrumentation	Object Oriented Programming
Microwave communication systems	MEMS	Human Computer Interface
Biomedical Electronic Systems	Analog VLSI Design	Mathematical Optimization
Digital Signal Processors	Electronic System Design	Augmented & Virtual Reality
Satellite Communication	VLSI Testing & Verification	Introduction to Game Design
RF Circuit Design	Internet of Things	Block-Chain and Ledger
Deep Learning	VLSI Packaging Aspects	Computer Ethics & Public Policy
Social Network Analysis	Robotics and Humanoids	Probabilistic Graphical Model
Wireless Sensor networks	Power electronics & renewable energy	Cyber Security
Advanced Internet Technologies	Distributed Computing Systems	Information Retrieval
Data Mining and Data Warehousing	Mobile Computing	Ethical Hacking
Bioinformatics		

- The Courses proposed by Industry experts will be offered with the involvement of a course coordinator nominated by the Institute.
- Any change in the elective courses shall be approved by the Senate.

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